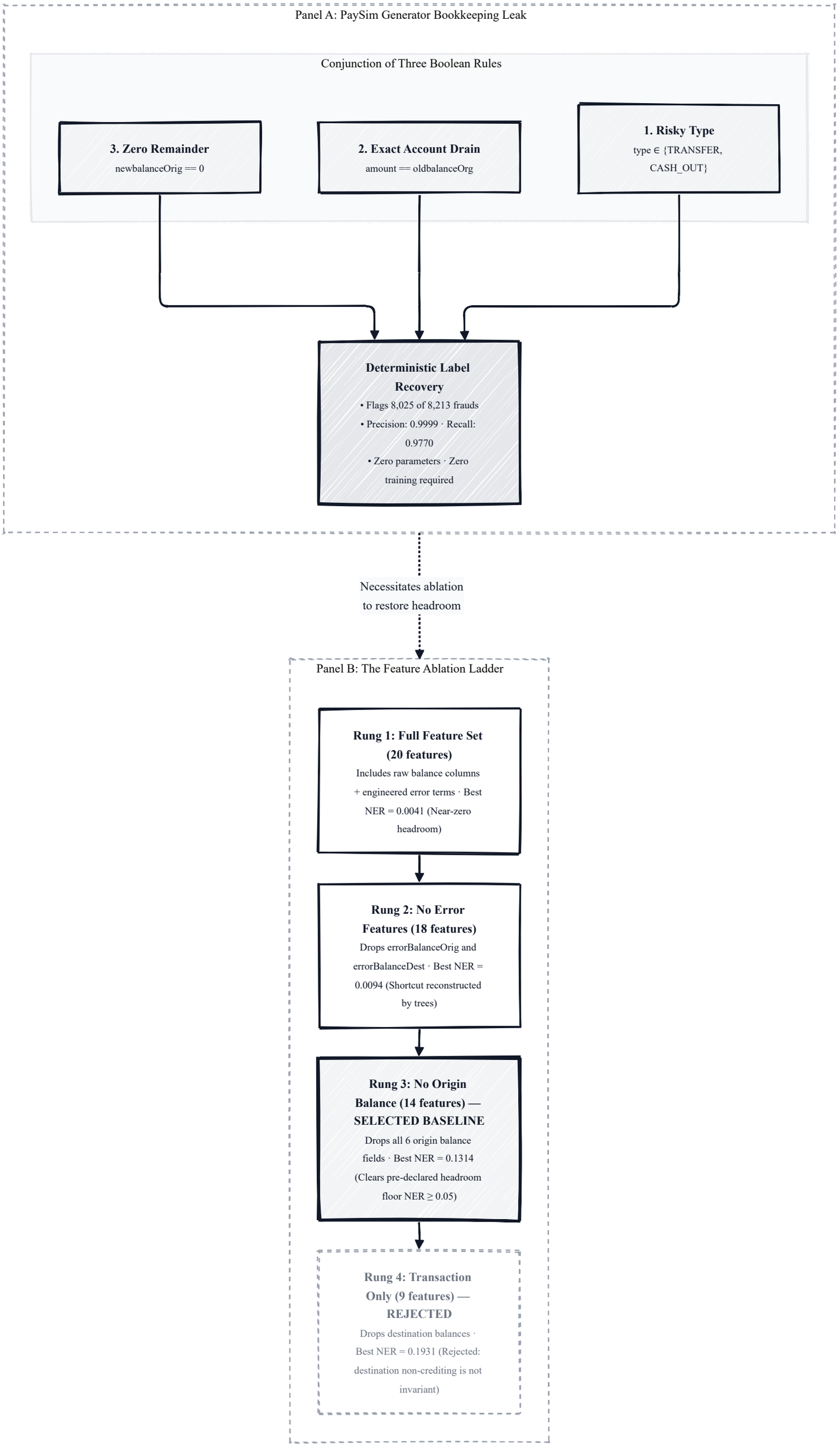

Panel A: PaySim Generator Bookkeeping Leak
Conjunction of Three Boolean Rules
3. Zero Remainder
newbalanceOrig == 0
2. Exact Account Drain
amount == oldbalanceOrg
1. Risky Type
type ∈ {TRANSFER, CASH_OUT}
Deterministic Label Recovery
• Flags 8,025 of 8,213 frauds
• Precision: 0.9999 · Recall: 0.9770
• Zero parameters · Zero training required
Necessitates ablation to restore headroom
Panel B: The Feature Ablation Ladder
Rung 1: Full Feature Set (20 features)
Includes raw balance columns + engineered error terms · Best NER = 0.0041 (Near-zero headroom)
Rung 2: No Error Features (18 features)
Drops errorBalanceOrig and errorBalanceDest · Best NER = 0.0094 (Shortcut reconstructed by trees)
Rung 3: No Origin Balance (14 features) — SELECTED BASELINE
Drops all 6 origin balance fields · Best NER = 0.1314 (Clears pre-declared headroom floor NER ≥ 0.05)
Rung 4: Transaction Only (9 features) — REJECTED
Drops destination balances · Best NER = 0.1931 (Rejected: destination non-crediting is not invariant)